

# Data Science

## Chapter 3: Probabilistic Models

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Data Science

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### Combinatorics

Introduction

Overview

Summary

### Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

### Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

### Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

- **Counting Combinatorics:** A branch of combinatorics that deals with forming possible arrangements from  $n$  elements.
- **Combinatorial Models:** Determine the number of possible arrangements.
- **Important considerations:**
  - Are all elements ( $k = n$ ) or only some ( $k < n$ ) of the elements used?
  - Are the elements distinguishable?
  - Is the order of elements relevant?
- There is the classical **urn model** in which all  $n$  elements are placed in one urn and  $k$  elements are drawn from it (a **sample**), and a generalized **balls-and-boxes** model, in which  $k$  balls are distributed among  $n$  boxes.
- Counting combinatorics forms the basis for calculating with **classical probabilities** (see later).

## Combinatorics

### Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

## Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

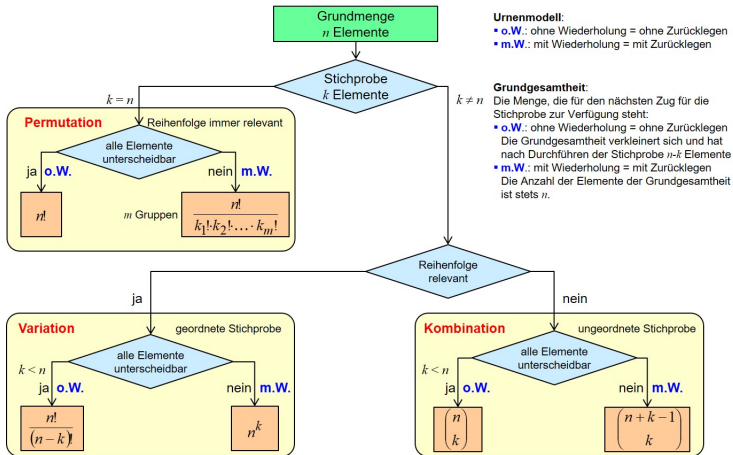
## Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

For the classical urn model, the numbers of possible arrangements can be determined according to the following overview:



# Summary

„Musical“ summary of the topic **Combinatorics (Maths-Song)**:

# KOMBINATORIK



[https://youtu.be/JoETgJS1oWE?si=YasBb9rD-8\\_q-GuH](https://youtu.be/JoETgJS1oWE?si=YasBb9rD-8_q-GuH)

## Combinatorics

Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

## Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

## Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

## Combinatorics

## Introduction

## Overview

## Summary

- ## Basics of Probability Theory

## Events, Sample Space & Event Set

- ## Operations with Events

## Probability

## Rules of Probability

## Conditional Probability & Stochastic Independence

## Bayes' Rule

### Definition

### Distribution Function

Discrete Random Variables

## Continuous Random Variables

Expected Value

### Variance

## Standardization

## Quantiles

## Preface

## Discrete Distributions

## Continuous Distributions

# Events, Sample Space & Event Set II

## Definition: Elementary Event

The individual outcomes of a random experiment that cannot be decomposed further and are mutually exclusive are called **elementary events**.

**Example** In a *die roll*, elementary outcomes are given by the number of pips on the top face. The experiment has six possible elementary events: 1, 2, 3, 4, 5, 6.

## Definition: Sample Space

The set  $\Omega$  of all elementary events of a random experiment is called the **event space**, **sample space**, or **basic set** of the random experiment.

**Example** In the random experiment *rolling a die* the sample space is  $\Omega = \{1, 2, 3, 4, 5, 6\}$ . The sample space contains a finite number of elements.

## Definition: Random Event

A **random event**  $A$  is a subset of the sample space  $\Omega$ . The (concrete) event  $A$  has occurred if the outcome of the random experiment is an element of this subset  $A$ .

**Example** In the experiment *rolling two dice*, consider the random event  $A = \{\text{sum of pips is greater than 10}\}$ . This event consists of three elementary outcomes:  $A = \{(5, 6), (6, 5), (6, 6)\}$ .

# Events, Sample Space & Event Set III

## Notation

The individual elementary events are often written in set notation  $\{*\}$  as follows:

$$\Omega = \{e_1, e_2, \dots, e_m, \dots\}$$

This notation can also be used for any (concrete) event  $A$ , e.g.,  
 $A = \{e_1, e_2, e_5\}$ :  $A$  contains the three elementary events  $e_1$ ,  $e_2$ , and  $e_5$ .

## Definition: Event Set

All events of a random experiment with sample space  $\Omega$  form the corresponding **event set**  $E(\Omega)$ .

# Events, Sample Space & Event Set IV

## Remarks on the Event Set $E(\Omega)$

All events means "the set of all subsets of the sample space  $\Omega$ ". In set theory, this is the power set of a base set (here, the sample space  $\Omega$ )<sup>1</sup>, i.e., a set whose elements are themselves sets.

Note that

- the **certain event**, the base set  $\Omega$ , belongs to  $E(\Omega)$ :  $\Omega \subseteq \Omega$ . Since it contains all elementary events, it always occurs whenever the random experiment is performed.
- the **impossible event**  $\emptyset$  belongs to  $E(\Omega)$ :  $\emptyset \subset \Omega$ . Since the empty set has no elements, this event never occurs.

For finite sample spaces  $\Omega$ , i.e.,  $|\Omega| = n \in \mathbb{N}_0$ , we have  $|E(\Omega)| = 2^n$ ; in general,  $|\Omega| < |E(\Omega)|$  (Cantor's theorem).

Examples:

- $\Omega = \emptyset \Rightarrow E(\Omega) = \{\emptyset\}$
- $\Omega = \{e\} \Rightarrow E(\Omega) = \{\emptyset, \{e\}\}$
- $\Omega = \{e_1, e_2\} \Rightarrow E(\Omega) = \{\emptyset, \{e_1\}, \{e_2\}, \{e_1, e_2\}\}$
- $\Omega = \{e_1, e_2, e_3\}$   
 $\Rightarrow E(\Omega) = \{\emptyset, \{e_1\}, \{e_2\}, \{e_3\}, \{e_1, e_2\}, \{e_1, e_3\}, \{e_2, e_3\}, \{e_1, e_2, e_3\}\}$

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<sup>1</sup>Also denoted by  $\mathcal{P}(\Omega)$



# Illustrative Examples

- Coin toss (heads, tails):  $\Omega_M = \{\text{heads}, \text{tails}\}$
- Die (number of pips):  $\Omega_W = \{1, 2, 3, 4, 5, 6\}$
- Coin & die:  $\Omega_{M \times W} = \{e_1 = \{\text{heads}, 1\}, e_2 = \{\text{heads}, 2\}, e_3 = \{\text{heads}, 3\}, e_4 = \{\text{heads}, 4\}, e_5 = \{\text{heads}, 5\}, e_6 = \{\text{heads}, 6\}, e_7 = \{\text{tails}, 1\}, e_8 = \{\text{tails}, 2\}, e_9 = \{\text{tails}, 3\}, e_{10} = \{\text{tails}, 4\}, e_{11} = \{\text{tails}, 5\}, e_{12} = \{\text{tails}, 6\}\}$

Examples of defining concrete events (from the subsets that can be formed from  $E(\Omega_{M \times W})$  with  $|E(\Omega_{M \times W})| = 2^{12} = 4.096$ ):

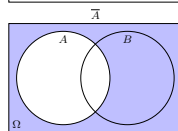
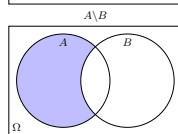
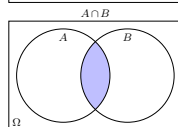
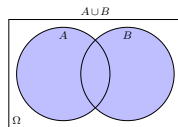
- $A = \{\text{number of pips} > 3\} = \{e_4, e_5, e_6, e_{10}, e_{11}, e_{12}\}$
- $B = \{\text{number of pips} < 6 \cap \text{heads}\} = \{e_7, e_8, e_9, e_{10}, e_{11}\}$
- $C = \{\text{number of pips} \leq 3 \cap \text{tails}\} = \{e_1, e_2, e_3\}$

Clearly,  $C = \overline{A \cup B} = \overline{A} \cap \overline{B}$  and  $A \cap C = \emptyset$ .

# Operations with Events I

Events are handled mathematically like sets:

- **Union**  $A \cup B$ : The event „**A or B**“ occurs exactly when  $A$ , or  $B$ , or both occur.
- **Intersection**  $A \cap B$ : The event „**A and B**“ occurs exactly when  $A$  and  $B$  occur simultaneously.
- **Difference**  $A \setminus B$ : The event „**A without B**“ occurs exactly when  $A$  occurs but  $B$  does not.
- **Complement**  $\bar{A}$ : The event „**not A**“ occurs exactly when  $A$  does not occur.



## Combinatorics

Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space & Event Set

## Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

## Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

## Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

# Operations with Events II

## Definition: Disjoint Events

Two events  $A$  and  $B$  are called **disjoint** or **mutually exclusive** if

$$A \cap B = \emptyset$$

## Definition: Complementary Event

The event

$$\overline{A} = \Omega \setminus A$$

is called the **complementary event** of  $A$ .

## Definition: Implied Event

If event  $A$  always occurs whenever event  $B$  occurs, i.e.,  $A \subset B$ , then we say that **event  $B$  implies event  $A$** .

**Note:** The cardinality  $|A|$  of a set is the number of elements contained in  $A$ .

# Probability

The possible events of a random experiment each have a certain chance of occurring.

It is therefore natural to assign a real number to every element of the event set  $E(\Omega)$  that **quantifies** this chance.

Such a mapping  $P$  is a real-valued function given by

$$\begin{aligned} P : E(\Omega) &\rightarrow \mathbb{R} \\ A &\mapsto P(A) \end{aligned}$$

This defines the probability  $P(A)$  of an event  $A$  in a formal way, but how exactly can probability be measured or specified, and which rules does it obey?

The answer is not entirely simple; there are

- 1 the „classical“ concept of probability,
- 2 the „axiomatic“ concept of probability.

The axiomatic concept of probability forms the foundation of probability theory and includes the classical concept.

## Combinatorics

Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

### Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

## Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

## Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

# Classical Probability

Laplace's view:

$$P(A) = \frac{\text{number of favorable outcomes}}{\text{number of possible outcomes}} = \frac{|A|}{|\Omega|}$$

A prerequisite for applying the classical concept of probability is that all elementary events  $e_1, e_2, \dots, e_n \subset \Omega$  are considered with equal probability:

$$P(e_1) = P(e_2) = \dots = P(e_n) = \frac{1}{n}$$

This concept is especially useful in games when determining the desired probability numerically. In this context, good knowledge of combinatorics is important for finding the counts  $|A|$  and  $|\Omega|$ .

## Combinatorics

Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

## Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

## Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

## Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

## Definition: Axioms of Probability Theory

Any function  $P$

$$\begin{aligned} P : E(\Omega) &\rightarrow \mathbb{R} \\ A &\mapsto P(A) \end{aligned}$$

that assigns a real number to each event  $A$  from the event set  $E(\Omega)$  is called a **probability function**, and the assigned real number  $P(A)$  is called the **probability** of  $A$ , provided that the following three Kolmogorov<sup>2</sup> axioms are satisfied:

**Axiom K1:**  $P(A) \geq 0 \quad \forall A \subset E$

The probability  $P(A)$  of every event  $A$  is a non-negative real number.

**Axiom K2:**  $P(\Omega) = 1$

The certain event  $\Omega$  has probability 1.

**Axiom K3:**  $P(A \cup B) = P(A) + P(B)$  if  $A \cap B = \emptyset$

Addition rule for disjoint events.

The tuple  $[\Omega, E, P]$  is called a probability space.

<sup>2</sup>Andrej Nikolaevič Kolmogorov, 1903–1987

# Rules of Probability I

Kolmogorov's axioms only specify the rules that a probability measure must satisfy.

## Theorem 1: Complementary Event

The probability of the event complementary to  $A$  is always

$$P(\overline{A}) = 1 - P(A), \quad \forall A \subset E$$

## Theorem 2: Impossible Event

The impossible event has probability

$$P(\emptyset) = 0$$

## Theorem 3: Pairwise Disjoint Events

If the events  $A_1, A_2, \dots, A_n \subset E$  are pairwise disjoint, then the probability of the event formed by the union of all these events equals the sum of the individual probabilities:

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = \sum_{j=1}^n P(A_j)$$

# Rules of Probability II

## Theorem 4: Difference Events

For a difference set  $A \setminus B$ , the following always holds:

$$P(A \setminus B) = P(A) - P(A \cap B)$$

## Theorem 5: Addition Rule

For any two events  $A$  and  $B$  from  $E$ , the following always holds:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

## Theorem 6: Monotonicity Property

If event  $A$  implies event  $B$ , then the probability of  $B$  is never smaller than the probability of  $A$ :

$$A \subset B \Rightarrow P(A) \leq P(B)$$

### Combinatorics

Introduction

Overview

Summary

### Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

### Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

### Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

### Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions



# Conditional Probability & Stochastic Independence I

The probability of occurrence of an event  $A$  „under the condition“ that event  $B$  has occurred (or occurs simultaneously with  $A$ ) is called the „conditional probability of  $A$  given  $B$ “ and is written as  $P(A|B)$ .

## Definition: Conditional Probability

Let  $A$  and  $B$  be two events of the probability space  $[\Omega, E, P]$ . The **conditional probability of  $A$  given  $B$**  is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

## Theorem 7: Multiplication Rule

The probability that two events  $A$  and  $B$  occur simultaneously is

$$P(A \cap B) = P(A|B) \cdot P(B) = P(B|A) \cdot P(A) = P(B \cap A)$$

# Conditional Probability & Stochastic Independence II

**Experiment:** A red and a green die are rolled (36 elementary events):

|       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|
| (1,1) | (1,2) | (1,3) | (1,4) | (1,5) | (1,6) |
| (2,1) | (2,2) | (2,3) | (2,4) | (2,5) | (2,6) |
| (3,1) | (3,2) | (3,3) | (3,4) | (3,5) | (3,6) |
| (4,1) | (4,2) | (4,3) | (4,4) | (4,5) | (4,6) |
| (5,1) | (5,2) | (5,3) | (5,4) | (5,5) | (5,6) |
| (6,1) | (6,2) | (6,3) | (6,4) | (6,5) | (6,6) |

**Events:**

- $A$ : “At least one die shows a 6”
- $B$ : “Sum of the dice/pips is greater than 9”

**Probabilities:**

$$P(A) = \frac{11}{36}, \quad P(B) = \frac{6}{36}, \quad P(A \cap B) = \frac{5}{36}$$

**Conditional Probability:**

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{5/36}{6/36} = \frac{5}{6}$$

**Interpretation:** Given that the sum exceeds 9, the probability that at least one die shows a six is  $\frac{5}{6}$ .

## Combinatorics

[Introduction](#)[Overview](#)[Summary](#)

## Basics of Probability Theory

[Events, Sample Space & Event Set](#)[Operations with Events](#)[Probability](#)[Rules of Probability](#)

## Conditional Probability & Stochastic Independence

[Bayes' Rule](#)

## Random variable

[Definition](#)[Distribution Function](#)[Discrete Random Variables](#)[Continuous Random Variables](#)[Expected Value](#)[Variance](#)[Standardization](#)[Quantiles](#)

## Stochastic Models & Selected Distributions

[Preface](#)[Discrete Distributions](#)[Continuous Distributions](#)

## Definition: Stochastic Independence

Two events  $A$  and  $B$  are called **(stochastically) independent** if

$$P(A|B) = P(A)$$

$$P(B|A) = P(B)$$

## Theorem 8: Multiplication Rule for Independent Events

If two events  $A$  and  $B$  are independent, then the probability that  $A$  and  $B$  occur simultaneously is exactly the product of the individual probabilities:

$$P(A \cap B) = P(A) \cdot P(B)$$

## Definition: Partition

Any  $n$  disjoint events  $H_1 \subset \Omega, H_2 \subset \Omega, \dots, H_n \subset \Omega$  that together cover the event space  $\Omega$  completely,

$$H_i \cap H_j = \emptyset \quad \forall i \neq j$$

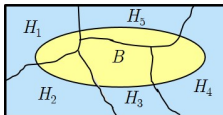
$$H_1 \cup H_2 \cup \dots \cup H_n = \Omega$$

are called a **partition** of  $\Omega$ .

## Theorem 9: Law of Total Probability

Let  $H_1, H_2, \dots, H_n$  be a partition of  $\Omega$ . Then for every  $B \subset E$ ,

$$P(B) = \sum_{j=1}^n P(B|H_j) \cdot P(H_j)$$



## Note

The term „total probability“ expresses that the probability of an event can be determined based on an arbitrarily chosen partition, provided that the conditional probabilities and the probabilities of the conditions are known.

### Combinatorics

Introduction

Overview

Summary

### Basics of Probability Theory

Events, Sample Space &amp; Event Set

Operations with Events

Probability

Rules of Probability

### Conditional Probability & Stochastic Independence

Bayes' Rule

### Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

### Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

## Example: Manufacturing

**Problem:** A product is manufactured by two machines:

- Machine  $M_1$ : produces  $\frac{2}{3}$  of all items, defect rate 10%
- Machine  $M_2$ : produces  $\frac{1}{3}$  of all items, defect rate 7%

**Given:**

- $P(\text{defect} \mid M_1) = 0.10, \quad P(M_1) = \frac{2}{3}$
- $P(\text{defect} \mid M_2) = 0.07, \quad P(M_2) = \frac{1}{3}$

**Law of Total Probability:**

$$P(\text{defect}) = P(\text{defect} \mid M_1) \cdot P(M_1) + P(\text{defect} \mid M_2) \cdot P(M_2)$$

**Calculation:**

$$P(\text{defect}) = 0.10 \cdot \frac{2}{3} + 0.07 \cdot \frac{1}{3} = 0.09$$

**Interpretation:** The overall probability that a randomly selected item is defective is 9%.

# Example: Three-Door Problem or Monty Hall Problem I

## The problem:

- Behind one of three doors there is a car; behind each of the others there is a goat.
- Hoping to win the car, the contestant first chooses one door.
- The host, who knows what is behind the doors, then opens another door with a goat behind it and offers the contestant the option to switch.

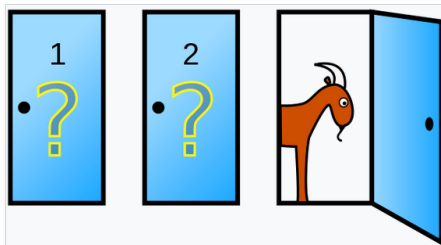


Abbildung: Wikipedia, *Monty Hall problem*, accessed on 02.05.2024

**Question:** Is it advantageous for the contestant to switch doors?

# Example: Three-Door Problem or Monty Hall Problem II

The door-choice events are defined as follows:

- $g_1$ : The contestant *initially* chose the door with the car
- $g_2$ : The contestant *initially* chose a door with a goat
- $G_1$ : The contestant wins the car *after switching*
- $G_2$ : The contestant wins a goat *after switching*

Initial event probabilities (without switching):

$$P(g_1) = \frac{1}{3} \text{ and } P(g_2) = \frac{2}{3}$$

Conditional event probabilities when switching:

$$P(G_1|g_1) = 0 \text{ and } P(G_1|g_2) = 1$$

Event probability when switching:

$$P(G_1) = P(G_1|g_1)P(g_1) + P(G_1|g_2)P(g_2) = 0 \cdot \frac{1}{3} + 1 \cdot \frac{2}{3} = \frac{2}{3}$$

# Bayes' Rule I

Bayes' rule establishes a relationship between two conditional probabilities in which the roles of condition and event are interchanged, e.g.,

$$P(A|B) \text{ and } P(B|A)$$

By rearranging Theorem 7, we obtain

$$\frac{P(A|B)}{P(B|A)} = \frac{P(A)}{P(B)}$$

and therefore

$$P(A|B) = \frac{P(A)}{P(B)} \cdot P(B|A) = \frac{P(B|A) \cdot P(A)}{P(B)} = \frac{P(B \cap A)}{P(B)}$$

## Note

The above relationship is often interpreted as follows:

- $A$  is an event whose probability  $P(A)$  is known before („a priori“) carrying out the random experiment.
- $B$  is an event that occurs in the outcome with probability  $P(B)$ .
- The relation above quantifies how the probability of  $A$  changes afterwards („a posteriori“) when it is known that  $B$  has actually occurred, i.e.,  $P(A) \mapsto P(A|B)$ .



## Bayes' Rule II

Bayes' rule also applies to an element  $H_i$  of a partition, and together with Theorem 9 this yields

$$P(H_i|B) = \frac{P(H_i) \cdot P(B|H_i)}{P(B)} = \frac{P(H_i) \cdot P(B|H_i)}{\sum_{j=1}^n P(B|H_j) \cdot P(H_j)}$$

from which Bayes' theorem follows directly:

## Bayes' Theorem

Let  $H_1, H_2, \dots, H_n$  be an arbitrary partition of  $\Omega$  and let  $B$  be an event with  $P(B) > 0$ . Then

$$P(H_i|B) = \frac{P(B|H_i) \cdot P(H_i)}{\sum_{j=1}^n P(B|H_j) \cdot P(H_j)}$$

## Example: Manufacturing

### Context (from previous example):

A product is manufactured by two machines:

- $M_1$ : produces  $\frac{2}{3}$  of items, defect rate 10%
- $M_2$ : produces  $\frac{1}{3}$  of items, defect rate 7%

**Question:** Given that a randomly selected item is defective, what is the probability that it was produced by  $M_1$ ?

**Recap:**

$$P(\text{defect}) = 0.09$$

**Bayes' Theorem:**

$$P(M_1 | \text{defect}) = \frac{P(\text{defect} | M_1) P(M_1)}{P(\text{defect})} = \frac{0.10 \cdot \frac{2}{3}}{0.09} = \frac{20}{27} \approx 0.741$$

**Interpretation:** If a product is defective, the probability that it was produced by  $M_1$  increases from 66.7% to about 74.1%.

# Example: Three-Door Problem or Monty Hall Problem

The following events are defined:

- $G_i$ : The car is behind door  $i$  ( $i = 1, 2, 3$ )
- $M_j$ : The host opened door  $j$  ( $j = 1, 2, 3$ )

The following situation is given:

The contestant chose door 1, and the host then opened door 3.

$\Rightarrow$  We are looking for the *conditional probability*  $P(G_2|M_3)$ .

The following probabilities are obtained:

- $P(G_1) = P(G_2) = P(G_3) = \frac{1}{3}$
- $P(M_3|G_1) = \frac{1}{2}$ ,  $P(M_3|G_2) = 1$ , and  $P(M_3|G_3) = 0$

Applying *Bayes' theorem* yields:

$$\begin{aligned} P(G_2|M_3) &= \frac{P(M_3|G_2)P(G_2)}{P(M_3|G_1)P(G_1) + P(M_3|G_2)P(G_2) + P(M_3|G_3)P(G_3)} \\ &= \frac{1 \cdot \frac{1}{3}}{\frac{1}{2} \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3}} = \frac{\frac{1}{3}}{\frac{1}{2} + \frac{1}{3}} = \frac{2}{5} \end{aligned}$$

**Consequence:** The contestant should switch in order to double the winning probability from initially  $\frac{1}{3}$  to now  $\frac{2}{3}$ .

# Definition of Random Variables

- Until now: Random events to which probabilities were assigned.
- From now on: Assign probabilities to real numbers: If an elementary event  $e \in \Omega$  occurs, then a real number  $\mathbf{X}(e)$  is assigned to this  $e$ .

## Definition: Random Variable

Given a probability space  $[\Omega, E, P]$ . A function

$$\begin{aligned}\mathbf{X} : \Omega &\rightarrow \mathbb{R} \\ e &\mapsto \mathbf{X}(e)\end{aligned}$$

that assigns a real number  $\mathbf{X}(e)$  to each elementary event  $e$  is called a **random variable** or **stochastic variable** if, for every real number  $x$ , there exists an event  $A_x \subset E$  with  $A_x = \{e | \mathbf{X}(e) \leq x\}$ .

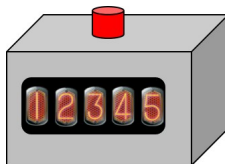
## Remarks:

- The random *variable*  $\mathbf{X}$  is a *function*.
- The probability function  $P$  is not needed for the definition.
- Introducing random variables largely eliminates the rather exhausting calculations with elementary events and events.

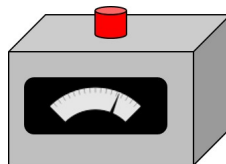
# Visualization of Random Variables

Less formally, this can be imagined as follows:

- Pressing the red button performs the random experiment inside the box
- The outcome, which is not visible from the outside, is the elementary event  $e$
- The display shows the associated real number  $\mathbf{X}(e)$
- Associated with this display value  $\mathbf{X}(e)$  is the event  $A_x = \{e | \mathbf{X}(e) \leq x\} \subset E$



discret



continuous

## Combinatorics

Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

## Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

## Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

# Example of a Random Variable

A coin is tossed once, so we consider the sample space

$$\Omega_M = \{\text{tails}, \text{heads}\}$$

and thus the event set  $E$  with four events

$$E = \{\emptyset, \{\text{tails}\}, \{\text{heads}\}, \Omega\}.$$

Let the random variable  $\mathbf{X} : \Omega_M \rightarrow \mathbb{R}$  denote the number of heads that can occur in this random experiment:

$$\mathbf{X}(\text{tails}) = x_1 = 0 \quad \text{and} \quad \mathbf{X}(\text{heads}) = x_2 = 1.$$

The defining condition “for all  $x \in \mathbb{R}$  there exists an event  $A_x \subset E$  with  $A_x = \{e | \mathbf{X}(e) \leq x\}$ ” is also satisfied:

$$\begin{aligned} -\infty < x < 0 : \quad A_x &= \emptyset, \\ 0 \leq x < 1 : \quad A_x &= \{\text{tails}\}, \\ 1 \leq x < \infty : \quad A_x &= \Omega. \end{aligned}$$

## Combinatorics

Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

## Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

## Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

# Distribution Function I

In addition to the real numbers  $x$  that a random variable  $\mathbf{X}$  can assume, the probabilities  $P$  with which these values occur are also of interest. Let

$P(\mathbf{X} = a) \Leftrightarrow$  probability that  $\mathbf{X} = a$

$P(a < \mathbf{X} \leq b) \Leftrightarrow$  probability that  $\mathbf{X}$  is greater than  $a$  and at most  $b$

$P(\mathbf{X} \leq x) \Leftrightarrow$  probability that  $\mathbf{X} \leq x$

Each of the three cases above can be uniquely associated with events and, through them, with the corresponding probabilities:

$$\begin{aligned}\mathbf{X} = a &\Leftrightarrow A = \{e | \mathbf{X}(e) = a\} \\ P(\mathbf{X} = a) &= P(A) = P(\{e | \mathbf{X}(e) = a\})\end{aligned}$$

Accordingly,

$$\begin{aligned}a < \mathbf{X} \leq b &\Leftrightarrow A = \{e | a < \mathbf{X}(e) \leq b\} \\ P(a < \mathbf{X} \leq b) &= P(A) = P(\{e | a < \mathbf{X}(e) \leq b\})\end{aligned}$$

Finally, for every real number  $x$ ,

$$\begin{aligned}\mathbf{X} \leq x &\Leftrightarrow A_x = \{e | \mathbf{X}(e) \leq x\} \\ P(\mathbf{X} \leq x) &= P(A_x) = P(\{e | \mathbf{X}(e) \leq x\})\end{aligned}$$

## Combinatorics

Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

## Random variable

Definition

### Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

## Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

## Definition: Distribution Function

The function

$$F(x) = P(\mathbf{X} \leq x) = P(A_x) = P(\{e | \mathbf{X}(e) \leq x\})$$

which assigns to each real number  $x$  the probability that the random variable  $\mathbf{X}$  takes on a value  $\mathbf{X} \leq x$ , is called the **distribution function** of the random variable  $\mathbf{X}$ . This probability is equal to the probability of the event  $A_x = \{e | \mathbf{X}(e) \leq x\}$ .

The distribution function has the following properties:

- ①  $F(x)$  is at least right-continuous at every point:

$$\lim_{\Delta x \rightarrow 0} F(x + \Delta x) = F(x), \quad \Delta x > 0$$

- ②  $F(x)$  is monotonically increasing everywhere:

$$F(a) \leq F(b), \quad \text{if } a < b, \quad a, b \in \mathbb{R}$$

- ③  $F(x)$  has the limits

$$\lim_{x \rightarrow -\infty} F(x) = 0, \quad \lim_{x \rightarrow +\infty} F(x) = 1$$

### Combinatorics

Introduction

Overview

Summary

### Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

### Random variable

Definition

#### Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

### Stochastic Models & Selected Distributions

Preface

Discrete Distributions

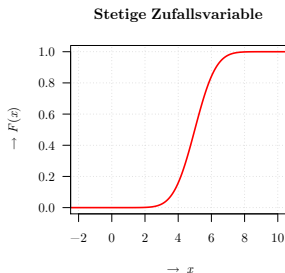
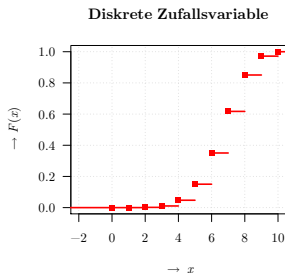
Continuous Distributions



# Distribution Function III

## Note

The properties of the distribution function are identical to those of the empirical distribution function!



## Combinatorics

Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

## Random variable

Definition

### Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

## Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

# Probability Mass Function I

In finite probability spaces, i.e.  $|\Omega| < \infty$ , the random variable is discrete, and there are  $n$  distinguishable real numbers  $x_i$ ,  $i = 1, \dots, n$  as realizations of  $\mathbf{X}$ .

## Definition: Probability Mass Function

If  $\mathbf{X}$  is a discrete random variable, then the function

$$f(x) = P(\mathbf{X} = x) = \begin{cases} P(\mathbf{X} = x_i) = p_i & \text{if } x = x_i, i = 1, \dots, n \\ 0 & \text{otherwise} \end{cases}$$

is called the **probability mass function** or simply the **mass function** of the random variable  $\mathbf{X}$  (sometimes also called the probability function).

### Properties

- $f(x) \geq 0$
- $\sum_{\text{all } i} f(x_i) = 1$
- $0 < f(x_i) \leq 1$

### Relationships

$$F(x) = P(\mathbf{X} \leq x) = \sum_{x_i \leq x} f(x_i)$$

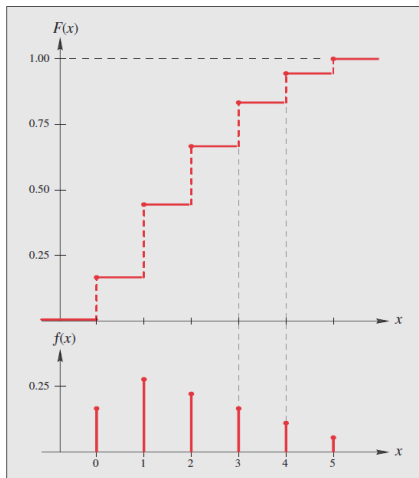
$$f(x) = F(x) - \lim_{\Delta x \rightarrow 0} F(x - \Delta x), \quad \Delta x > 0$$

### Note

The distribution function  $F(x)$  of discrete random variables is not differentiable at the jump points  $x = x_i$ !

# Probability Mass Function II

Relationship between distribution function  $F(x)$  and probability mass function  $f(x)$  of a discrete random variable  $\mathbf{X}$  [Schira, Statistische Methoden der VWL und BWL, 2021]:



## Combinatorics

Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

## Random variable

Definition

Distribution Function

### Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

## Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

# Density Function I

In infinite probability spaces, the random variable is continuous. The realizations of  $\mathbf{X}$  typically lie in intervals of the real numbers, for example in the interval  $[a, b]$ ,  $a, b \in \mathbb{R}$ .

## Definition: Probability Density Function

If  $\mathbf{X}$  is a continuous random variable with distribution function  $F(x)$ , then the first derivative of  $F(x)$  with respect to  $x$

$$f(x) = \frac{dF(x)}{dx}$$

is called the **probability density function**, or simply the **density function**, of the random variable  $\mathbf{X}$ .

## Properties

$$(1) \quad f(x) \geq 0$$

$$(2) \quad \int_{-\infty}^{+\infty} f(x) dx = 1$$

## Note

In the case of continuous random variables,  $F(x)$  is in general everywhere differentiable, and, hence,  $f(x)$  is everywhere defined.

# Density Function II

Remarks:

- ① The density function  $f(x)$  describes the **increase in probability** or the **probability rate** at the point  $x$ . Therefore, only the **area** enclosed by  $f(x)$  and the  $x$ -axis over a given interval is a probability:

$$P(a < \mathbf{X} \leq b) = \int_a^b f(x) dx = F(b) - F(a)$$

$$P(\mathbf{X} \leq x) = \int_{-\infty}^x f(u) du$$

- ② The interval  $[a, b]$  can be decomposed into three disjoint intervals:  $[a] \cup (a, b) \cup [b]$ . For a finite continuous  $f(x)$ , the following holds:

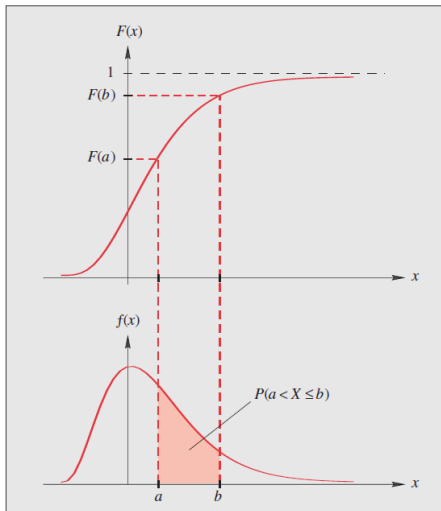
$$P(a \leq \mathbf{X} \leq b) = \underbrace{\int_a^a f(x) dx}_{P(\mathbf{X}=a)=0} + \underbrace{\int_a^b f(x) dx}_{P(a < \mathbf{X} < b)} + \underbrace{\int_b^b f(x) dx}_{P(\mathbf{X}=b)=0}$$

and therefore

$$P(a \leq \mathbf{X} \leq b) = P(a < \mathbf{X} \leq b) = P(a \leq \mathbf{X} < b) = P(a < \mathbf{X} < b)$$

# Density Function III

Probability as area below the density function  $f(x)$  [Schira, Statistische Methoden der VWL und BWL, 2021]:



## Combinatorics

Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

## Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

## Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

# Expected Value of a Random Variable I

The expected value is comparable to the empirical arithmetic mean. It describes the number that the random variable  $\mathbf{X}$  takes „on average“ (a measure of location)  $\rightarrow$  a number that does not necessarily have to be a realization of  $\mathbf{X}$ .

## Definition: Expected Value of a Random Variable

Let  $\mathbf{X}$  be a random variable and  $f(x)$  its mass or density function. Its **expected value** is defined as

$$E(\mathbf{X}) = \sum_{\text{all } j} x_j \cdot f(x_j) \quad \text{discrete}$$

$$E(\mathbf{X}) = \int_{-\infty}^{+\infty} x \cdot f(x) dx \quad \text{continuous}$$

## Notation:

The expected value is also called the **mean** or **mathematical expectation**. As a formula symbol for the resulting number, one usually uses  $\mu$ ,  $\mu_{\mathbf{X}}$ , or  $\mu(\mathbf{X})$ .

# Expected Value of a Random Variable II

## Example A (discrete random variable)

Let the **discrete** random variable **X** have the following **probability mass function**:

$$f(x) = \begin{cases} \frac{1}{3}, & \text{for } x = 1, \\ \frac{2}{3}, & \text{for } x = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Then the expected value is

$$E(\mathbf{X}) = 1 \cdot \frac{1}{3} + 2 \cdot \frac{2}{3} = \frac{5}{3} \approx 1.666\dots$$

### Combinatorics

Introduction

Overview

Summary

### Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

### Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

### Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions



# Expected Value of a Random Variable III

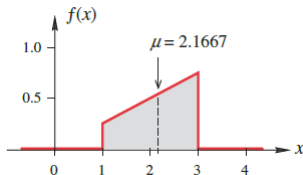
## Example B (continuous variable)

Let the **continuous** random variable **X** have the following **density function**:

$$f(x) = \begin{cases} \frac{1}{4}x, & \text{for } 1 \leq x \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Then the expected value is:

$$\begin{aligned} \mu = E(\mathbf{X}) &= \int_{-\infty}^{\infty} x \cdot f(x) dx = \int_{-\infty}^1 0 dx + \int_1^3 x \cdot \frac{1}{4}x dx + \int_3^{\infty} 0 dx \\ &= \int_1^3 \frac{1}{4}x^2 dx = \left[ \frac{1}{12}x^3 \right]_1^3 = \frac{27}{12} - \frac{1}{12} = \frac{26}{12} = 2.1667. \end{aligned}$$



# Expected Value of Transformation $Y = g(X)$ I

Let  $Y = g(X)$  be the transformation of the random variable  $X$ . Then it holds:

$$E(Y = g(X)) = \sum_{\text{all } j} g(x_j) \cdot f(x_j) \quad \text{discrete}$$

$$E(Y = g(X)) = \int_{-\infty}^{+\infty} g(x) \cdot f(x) dx \quad \text{continuous}$$

The following rules apply for **calculations with expected values**:

- ① **Constant:** For  $y = g(x) = a$ , it follows that

$$E(Y) = E(a) = a$$

- ② **Factor:** For  $y = b \cdot g(x)$ , it follows that

$$E(Y) = E(b \cdot g(X)) = b \cdot E(g(X))$$

- ③ **Linear Transformation:** For  $y = a + b \cdot x$ , it follows that

$$E(Y) = E(a + b \cdot X) = a + b \cdot E(X)$$

- ④ **Linear Combination:** For  $y = \lambda_1 \cdot g_1(x) + \lambda_2 \cdot g_2(x)$ , it follows that

$$E(Y) = E(\lambda_1 \cdot g_1(X) + \lambda_2 \cdot g_2(X)) = \lambda_1 \cdot E(g_1(X)) + \lambda_2 \cdot E(g_2(X))$$

# Expected Value of Transformation $Y = g(X)$ II

**Exemplary Problem:** Failures occur during the operation of a production facility. Based on past observations, the following probability distribution is assumed for the number of failures per day  $X$ :

|                   |      |      |      |      |
|-------------------|------|------|------|------|
| $x$               | 0    | 1    | 2    | 3    |
| $f(x) = P(X = x)$ | 0.35 | 0.40 | 0.15 | 0.10 |

Resolving failures follow the degressive cost function (in thousand Euro):

$$g(x) = 5 - \frac{4}{x+1}$$

**Substituting the expectation of failures**,  $E(X) = 1$  into the cost function yields expected costs of  $g(E(X)) = 5 - \frac{4}{2} = 3$ .  $\rightarrow$  daily costs of 3000 Euro

**Compute transformed values**

|        |   |   |      |   |
|--------|---|---|------|---|
| $x$    | 0 | 1 | 2    | 3 |
| $g(x)$ | 1 | 3 | 3.66 | 4 |

**Compute expectation**

$$E(g(X)) = \sum_x g(x) \cdot f(x) = 1 \cdot 0.35 + 3 \cdot 0.40 + 3.66 \cdot 0.15 + 4 \cdot 0.10 = 2.5$$

**Conclusion:**

Lower expected daily costs of 2500 Euro since  $E(g(X)) \neq g(E(X))$ .

# Variance of a Random Variable I

The variance corresponds to the empirical variance and represents a *measure of dispersion around the mean*.

## Definition: Variance of the Random Variable

Let  $\mathbf{X}$  be a random variable and  $\mu_X$  its expected value. The expected value of the squared deviations of the random variable from its expected value

$$\text{Var}(\mathbf{X}) = E((\mathbf{X} - \mu_X)^2)$$

is called the **variance** of the random variable  $\mathbf{X}$ , provided that the sum or the integral exists. The positive square root of the variance

$$\sigma_X = +\sqrt{\text{Var}(\mathbf{X})}$$

is called the **standard deviation**.

## Notation:

As a formula symbol, especially for the resulting number,  $\sigma^2$ ,  $\sigma_X^2$ , or  $\sigma^2(\mathbf{X})$  is usually used for  $\text{Var}(\mathbf{X})$ .

## Note:

The variance is the expected value of the transformation  $y = g(x) = (x - \mu_X)^2$ .

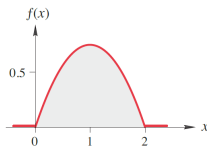
# Variance of a Random Variable II

According to the definition, the variance is calculated as follows:

$$\text{Var}(\mathbf{X}) = \sum_{\text{all } j} (x_j - \mu_X)^2 \cdot f(x_j) \quad \text{discrete}$$

$$\text{Var}(\mathbf{X}) = \int_{-\infty}^{+\infty} (x - \mu_X)^2 \cdot f(x) dx \quad \text{continuous}$$

**Example:** Given  $\mathbf{X}$  with density  $f(x) = \frac{3}{2} \left(x - \frac{1}{2}x^2\right)$  in the interval  $[0, 2]$ :



According to the symmetry, the expected value is  $\mu_X = E(\mathbf{X}) = 1$ .  
The variance is computed via the definite integral

$$\text{Var}(\mathbf{X}) = \int_0^2 (x-1)^2 \cdot \frac{3}{2} \left(x - \frac{1}{2}x^2\right) dx = \dots = \frac{1}{5}$$

and the standard deviation yields  $\sigma_X = 0.4472$ .

## Combinatorics

Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

## Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

## Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

# Variance of a Random Variable III

The following rules apply for **calculations with variances**:

- ① **Simplified Calculation**: The following holds (see item 6 for  $d = 0$ ):

$$\text{Var}(\mathbf{Y}) = \text{E}(\mathbf{X}^2) - \mu_X^2$$

- ② **Constant**: Since a constant quantity does not vary, the following holds:

$$\text{Var}(a) = 0$$

- ③ **Shift**: For  $y = x + a$ , it follows that

$$\text{Var}(\mathbf{Y}) = \text{Var}(\mathbf{X} + a) = \text{Var}(\mathbf{X})$$

- ④ **Scaling**: For  $y = b \cdot x$ , it follows that

$$\text{Var}(\mathbf{Y}) = \text{Var}(b \cdot \mathbf{X}) = b^2 \cdot \text{Var}(\mathbf{X})$$

- ⑤ **Linear Transformation**: For  $y = a + b \cdot x$ , it follows that

$$\text{Var}(\mathbf{Y}) = \text{Var}(a + b \cdot (\mathbf{X})) = b^2 \cdot \text{Var}(\mathbf{X})$$

- ⑥ **Steiner's Translation Theorem**: For  $d = \text{const.}$ , the following holds:

$$\text{Var}(\mathbf{Y}) = \text{E}((\mathbf{X} - d)^2) - (\mu_X - d)^2$$

## Combinatorics

Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

## Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

## Variance

Standardization

Quantiles

## Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

# Standardization of a Random Variable

Every random variable can be transformed in such a way that the resulting random variable has **zero mean** and its **variance equals one**.

For the purpose of comparability, this is often desirable or necessary.

## Definition: Standardization

If  $\mathbf{X}$  is a random variable with expected value  $\mu$  and standard deviation  $\sigma > 0$ , then the linearly transformed random variable

$$\mathbf{Z} = \frac{\mathbf{X} - \mu}{\sigma} = -\frac{\mu}{\sigma} + \frac{1}{\sigma} \cdot \mathbf{X} = a + b \cdot \mathbf{X}$$

is called **standardized**.

## Remarks

- ① As required (see calculation rules), the following holds:

$$\begin{aligned} E(\mathbf{Z}) &= a + b \cdot E(\mathbf{X}) = -\frac{\mu}{\sigma} + \frac{1}{\sigma} \cdot \mu = 0 \\ \sqrt{\text{Var}(\mathbf{Z})} &= |b| \cdot \sqrt{\text{Var}(\mathbf{X})} = \frac{1}{\sigma} \cdot \sigma = 1 \end{aligned}$$

- ② Intuitively, the linear transformation  $\mathbf{Z} = g(\mathbf{X}) = \frac{\mathbf{X} - \mu}{\sigma}$  first represents a **shift**  $\mathbf{Y} = \mathbf{X} - \mu$  by  $\mu$ , and then a **scaling**  $\mathbf{Z} = \frac{\mathbf{Y}}{\sigma}$  by the factor  $\frac{1}{\sigma}$ .

Just as in the case of empirical frequencies, the quantiles of a distribution function  $F(x)$  can be defined with the same interpretation.

## Definition: Quantile

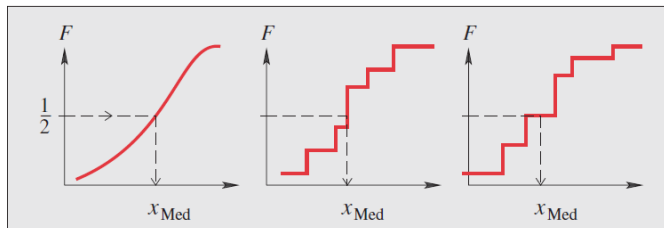
The number  $x_{[q]}$  for which simultaneously

$$P(\mathbf{X} \leq x_{[q]}) \geq q \quad \text{and} \quad P(\mathbf{X} \geq x_{[q]}) \geq 1 - q$$

holds is called the  $q$ -quantile of the distribution.

## Remember:

For  $q = 0.5$ ,  $x_{[q]} = x_{\text{Med}}$  is called the *Median* with  $x_{\text{Med}} = F^{-1}(0.5)$ :





## Preliminary Remarks on Selected Distributions

**So far:** random variable as a mathematical (abstract) concept.

**Now:** apply this concept to quantitatively describe a **random process** or **random experiment** by specifying a concrete distribution function.

## Notes

For some distributions, so-called “**stochastic models**” can be given that follow this distribution (e.g., urn models). A concrete random process is uniquely specified only after choosing numerical values for the **distribution parameters**.

The distribution functions presented here are selected according to **two criteria**:

- ① **Fundamental distributions** that should be part of basic statistical literacy and for which simple model interpretations usually exist.
- ② **Relevance for inferential statistics.** In that context, distribution functions are also required for which no stochastic model exists.

### Events, Sample Space & Event Set

## Operations with Events

## Probability

## Rules of Probability

## Conditional Probability & Stochastic Independence

### Bayes' Rule

## Random variable

### Distribution Function

Discrete Random Variables

## Continuous Random Variables

Expected Value

### Variance

### Standardization

### Quantiles

## Stochastic Models & Selected Distributions

Discrete Distributions

Continuous Distributions

## Model

The discrete random variable  $\mathbf{X}_U$  has  $m$  different realizations  $x_1, x_2, \dots, x_m$ , all of which have the same probability (*Laplace experiment*).

The distances between individual realizations do not need to be equal, although this is often the case.

## Uniform Distribution

If the realizations of a uniformly distributed random variable are the natural numbers from 1 to  $m$ , then:

$$f_U(x; m) = \begin{cases} \frac{1}{m} & \text{for } x = 1, 2, \dots, m \\ 0 & \text{otherwise} \end{cases}$$

$$E(\mathbf{X}_U) = \frac{m+1}{2}$$

$$\text{Var}(\mathbf{X}_U) = \frac{m^2 - 1}{12}$$

## Combinatorics

Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space &amp; Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability &amp; Stochastic Independence

Bayes' Rule

## Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

## Stochastic Models &amp; Selected Distributions

Preface

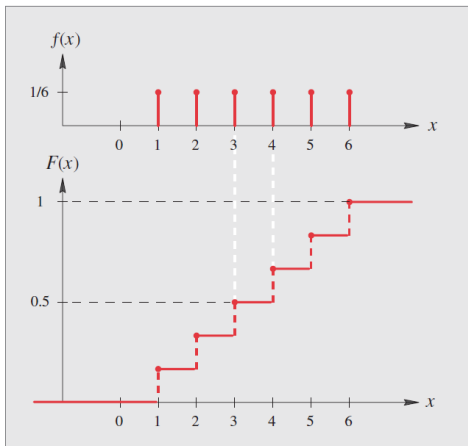
Discrete Distributions

Continuous Distributions

# Uniform Distribution II

## Example

For  $m = 6$ , we can consider the classical *Laplace experiment* „rolling a die“ for the random variable  $\mathbf{X}_U$ : „number of pips“ with the following probability mass function  $f(x)$  and distribution function  $F(x)$  [Schira, Statistische Methoden der VWL und BWL, 2021]:



## Model

The discrete random variable  $\mathbf{X}_{\text{Be}}$  has the two realizations 0 and 1. The random experiment has only the events  $A$  ("success", the event of interest occurs) and  $\bar{A}$  ("failure", the event of interest does not occur). The probability of  $A$  is  $P(A) = p$ .

## Bernoulli Distribution

$$f_{\text{Be}}(x; p) = \begin{cases} 1 - p & \text{for } x = 0 \\ p & \text{for } x = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$E(\mathbf{X}_{\text{Be}}) = p$$

$$\text{Var}(\mathbf{X}_{\text{Be}}) = p \cdot (1 - p)$$

## Note

The Bernoulli distribution can be interpreted as the so-called indicator function of the set  $A \subset E$ . Its expected value equals the probability of the event  $A$ .

### Combinatorics

Introduction

Overview

Summary

### Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

### Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

### Stochastic Models & Selected Distributions

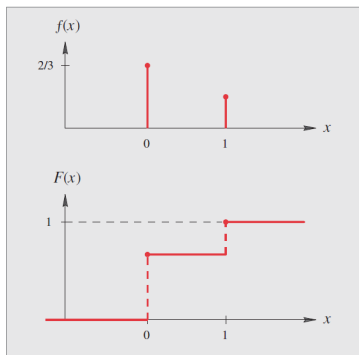
Preface

Discrete Distributions

Continuous Distributions

## Example

- 1 To be able to leave the starting area in *Ludo*, one must roll at least one six in three attempts ("success").  
The probability of success is  $p = \frac{91}{216} \approx 0.4213$ . Thus, the expected value is 0.4213, and the variance is  $pq = 0.4213 \cdot 0.5787 = 0.2438$ .
- 2 For a Bernoulli distributed random variable  $\mathbf{X}_{\text{Be}}$  with  $p = \frac{1}{3}$ , we obtain the following probability mass function  $f(x)$  and distribution function  $F(x)$  [Schira, Statistische Methoden der VWL und BWL, 2021]:



## Combinatorics

Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space &amp; Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability &amp; Stochastic Independence

Bayes' Rule

## Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

## Stochastic Models &amp; Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

## Model

Urn model **with** replacement:  $N$  balls, of which  $0 \leq S \leq N$  are black. The discrete random variable  $\mathbf{X}_{\text{Bi}}$  gives the number of black balls drawn in  $n$  draws with replacement. Therefore, in every draw the probability of drawing a black ball is constant and equals  $p = \frac{S}{N}$ .

## Binomial Distribution

With the binomial coefficient  $\binom{n}{x} = \frac{n!}{x! \cdot (n-x)!}$ , we have

$$f_{\text{Bi}}(x; p, n) = \binom{n}{x} \cdot p^x \cdot (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(\mathbf{X}_{\text{Bi}}) = n \cdot p$$

$$\text{Var}(\mathbf{X}_{\text{Bi}}) = n \cdot p \cdot (1-p)$$

## Remark

A binomially distributed random variable can also be interpreted as the sum of  $n$  independent Bernoulli-distributed random variables  $\mathbf{X}_{\text{Be},1}, \dots, \mathbf{X}_{\text{Be},n}$ , i.e.,

$$\mathbf{X}_{\text{Bi}} = \sum_{i=1}^n \mathbf{X}_{\text{Be},i}.$$

## Combinatorics

Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space &amp; Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability &amp; Stochastic Independence

Bayes' Rule

## Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

## Stochastic Models &amp; Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

# Binomial Distribution II

## Example (continued)

To be able to leave the starting area in *Ludo*, one must roll at least one six in three attempts (“success”).

We have already stated that the probability of success is  $p = \frac{91}{216} \approx 0.4213$ . We want to verify this.

To do so, we add the probabilities of the three disjoint events

$$P(\mathbf{X}_{Bi} = 1) + P(\mathbf{X}_{Bi} = 2) + P(\mathbf{X}_{Bi} = 3)$$

with  $p = \frac{1}{6}$ . Rolling at least one six means rolling either one, two, or three sixes.

We compute:

$$\binom{3}{1} \frac{1}{6} \left(\frac{5}{6}\right)^2 + \binom{3}{2} \left(\frac{1}{6}\right)^2 \frac{5}{6} + \binom{3}{3} \left(\frac{1}{6}\right)^3 = (3 \cdot 25 + 3 \cdot 5 + 1) \frac{1}{216} = \frac{91}{216}.$$

### Combinatorics

Introduction

Overview

Summary

### Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

### Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

### Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

## Model

Limit case of the discrete binomial distribution for a “huge number” of trials ( $n \rightarrow \infty$ ) together with a “very small” success probability ( $p \rightarrow 0$ ), while  $\lambda = n \cdot p = \text{const.} \in \mathbb{R}^+$ .

## Poisson Distribution

$$\begin{aligned}f_{\text{Po}}(x; \lambda) &= \frac{\lambda^x}{x!} \cdot e^{-\lambda}, \quad x = 0, 1, \dots, n \\E(\mathbf{X}_{\text{Po}}) &= \lambda \\ \text{Var}(\mathbf{X}_{\text{Po}}) &= \lambda\end{aligned}$$

## Remark

- Under the conditions  $n$  “large” and  $p$  “small” with  $0 < \lambda = n \cdot p < \infty$ , the Poisson distribution is a good approximation of the exact binomial distribution.
- This approximation improves with increasing  $n$  increases and decreasing  $p$ .

### Combinatorics

Introduction

Overview

Summary

### Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

### Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

### Stochastic Models & Selected Distributions

Preface

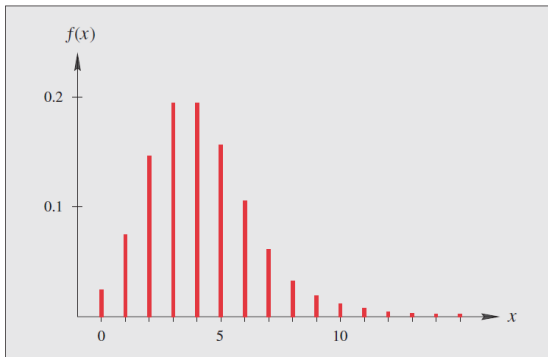
Discrete Distributions

Continuous Distributions



## Example

- Number of typos per page, or lightning strikes per hectare per year, ...
- For a Poisson distributed random variable  $X_{Po}$  with  $\lambda = 4$ , we obtain the following probability mass function  $f(x)$  [Schira, Statistische Methoden der VWL und BWL, 2021]:



### Combinatorics

Introduction

Overview

Summary

### Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

### Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

### Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

# Poisson Distribution III

## Example

### Problem:

A player repeatedly bets on the number 17 in roulette.

What is the probability that he wins exactly 8 times in 200 games?

### Modeling:

- Probability of success in one game:  $p = \frac{1}{37}$
- Number of trials:  $n = 200$
- Expected value:  $\lambda = np = \frac{200}{37} \approx 5.4054$

### Poisson Approximation:

$$P(\mathbf{X}_{\text{Po}} = 8) = f_{\text{Po}}(8; 5.4054) = \frac{\lambda^8}{8!} e^{-\lambda} = \frac{5.4054^8}{8!} e^{-5.4054} \approx 0.0812$$

### Comparison (Binomial):

$$P(\mathbf{X}_{\text{Bi}} = 8) = f_{\text{Bi}}\left(8; \frac{1}{37}; 200\right) = \binom{200}{8} \left(\frac{1}{37}\right)^8 \left(\frac{36}{37}\right)^{192} \approx 0.0814$$

### Conclusion:

The Poisson approximation is very accurate for large  $n$  and small  $p$ .

## Model

A Bernoulli experiment with success probability  $p$  is repeated independently until the first success occurs. The discrete random variable  $\mathbf{X}_{\text{Geo}} \in \mathbb{N}_0$  gives the number of **preceding failures**.

## Geometric Distribution

$$f_{\text{Geo}}(x; p) = (1 - p)^x \cdot p, \quad x = 0, 1, 2, \dots$$

$$F_{\text{Geo}}(x; p) = 1 - (1 - p)^{x+1}$$

$$\mathbb{E}(\mathbf{X}_{\text{Geo}}) = \frac{1 - p}{p}$$

$$\text{Var}(\mathbf{X}_{\text{Geo}}) = \frac{1 - p}{p^2}$$

## Note

In the literature, an alternative model is also used in which the geometrically distributed random variable  $\mathbf{Y}_{\text{Geo}}$  gives the number of trials **up to the first success**, i.e.,  $\mathbf{Y}_{\text{Geo}} \in \mathbb{N}$ . Trivially,  $\mathbf{Y}_{\text{Geo}} = \mathbf{X}_{\text{Geo}} + 1$ .

## Combinatorics

Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space &amp; Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability &amp; Stochastic Independence

Bayes' Rule

## Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

## Stochastic Models &amp; Selected Distributions

Preface

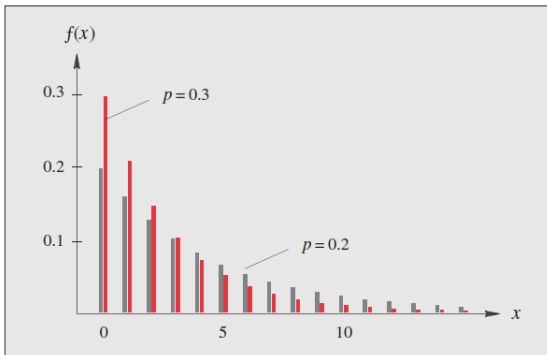
Discrete Distributions

Continuous Distributions

# Geometric Distribution II

## Example

For Geometric distributed random variables  $X_{\text{Geo}}$  with  $p = 0.2$  and  $p = 0.3$ , we obtain the following probability mass functions  $f(x)$  [Schira, Statistische Methoden der VWL und BWL, 2021]:



### Combinatorics

Introduction

Overview

Summary

### Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

### Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

### Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

## Example

- In Tunis, hailing a taxi is not guaranteed; waiting time is acceptable due to many taxis.
- Probability that a taxi arrives within the next minute:  $p = 0.2$ .
- Let  $X_{\text{Geo}}$  be the number of unsuccessful minutes waited (number of failures before the first success).

| $x$                      | 0   | 1    | 2     | 3      | 4      | 5      | ... |
|--------------------------|-----|------|-------|--------|--------|--------|-----|
| $f_{\text{Geo}}(x; 0.2)$ | 0.2 | 0.16 | 0.128 | 0.1024 | 0.0819 | 0.0655 | ... |
| $F_{\text{Geo}}(x; 0.2)$ | 0.2 | 0.36 | 0.488 | 0.5904 | 0.6723 | 0.7378 | ... |

The probability of having to wait no longer than three minutes is

$$F_{\text{Geo}}(2; 0.2) = 0.488.$$

The probability of having to wait at least five minutes is

$$P(X_{\text{Geo}} > 5) = 1 - F_{\text{Geo}}(5; 0.2) = 0.2622.$$

The random variable has expected value  $E(X_{\text{Geo}}) = \frac{q}{p} = \frac{0.8}{0.2} = 4$ , which means that, on average, one has to wait four full minutes without success.

## Model

Continuous version of the *Geometric distribution*. The curve of the strictly monotonically decreasing exponential function  $e^{-x}$  encloses, together with the  $x$ - and  $y$ -axes in the first quadrant, an area of size 1 area unit and is therefore suitable as a *density function*.

## Exponential Distribution

$$\begin{aligned}f_{\text{Ex}}(x; \lambda) &= \lambda \cdot e^{-\lambda x}; \quad x \geq 0; \lambda > 0 \\F_{\text{Ex}}(x) &= 1 - e^{-\lambda x} \\E(\mathbf{X}_{\text{Ex}}) &= \frac{1}{\lambda} \\\text{Var}(\mathbf{X}_{\text{Ex}}) &= \frac{1}{\lambda^2}\end{aligned}$$

## Note

The exponential distribution is a *memoryless distribution*: let  $w$  be an already elapsed waiting time, then:

$$f_{\text{Ex}|w}(w+x \mid X > w; \lambda) = f_{\text{Ex}}(x; \lambda)$$

## Combinatorics

Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space &amp; Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability &amp; Stochastic Independence

Bayes' Rule

## Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

## Stochastic Models &amp; Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

# Rectangular Distribution

## Model

A continuous random variable  $\mathbf{X}_R$  whose realizations come from an interval  $[a, b]$ .  $\mathbf{X}_R$  has the property that the probability of obtaining a result in an interval  $\Delta x \subseteq [a, b]$  is proportional to the length of this interval, independent of the interval's position within  $[a, b]$ . This is the continuous version of the *Uniform distribution*.

## Rectangular Distribution

$$f_R(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

$$F_R(x) = \begin{cases} 0 & \text{for } -\infty \leq x < a \\ \frac{x-a}{b-a} & \text{for } a \leq x \leq b \\ 1 & \text{for } b < x \leq +\infty \end{cases}$$

$$E(\mathbf{X}_R) = \frac{a+b}{2}$$

$$\text{Var}(\mathbf{X}_R) = \frac{(b-a)^2}{12}$$

## Combinatorics

Introduction

Overview

Summary

## Basics of Probability Theory

Events, Sample Space &amp; Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability &amp; Stochastic Independence

Bayes' Rule

## Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

## Stochastic Models &amp; Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

## Model

Limit case of the **standardized** *Binomial Distribution* for  $n \rightarrow \infty$ .

## Standard Normal Distribution

A continuous random variable **Z** with density function

$$f_{\text{St}}(z) = \frac{1}{\sqrt{2 \cdot \pi}} e^{-\frac{1}{2} z^2}$$

is called **standard normally distributed**.

## Properties

- 1 Maximum at  $z = 0$
- 2 Symmetry with respect to the axis  $z \equiv 0$
- 3 Inflection points at  $z = \pm 1$
- 4 We always have  $f_{\text{St}}(z) > 0$  (i.e.,  $f_{\text{St}}(z)$  has no zeros)
- 5 In the limit:  $\lim_{z \rightarrow \pm \infty} f_{\text{St}}(z) = 0$

### Combinatorics

Introduction

Overview

Summary

### Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

### Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

### Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions



## 5 The distribution function

$$F_{\text{St}}(z) = \frac{1}{\sqrt{2 \cdot \pi}} \int_{-\infty}^z e^{-\frac{1}{2}u^2} du$$

is **not available in closed form analytically**. Its values are tabulated and can also be computed via function calls.

## 6 Areas:

$$P(\mathbf{Z} \leq z) = \int_{-\infty}^z f_{\text{St}}(u) du = F_{\text{St}}(z)$$

$$P(a < \mathbf{Z} \leq b) = \int_a^b f_{\text{St}}(z) dz = F_{\text{St}}(b) - F_{\text{St}}(a)$$

$$P(-a < \mathbf{Z} \leq a) = \int_{-a}^a f_{\text{St}}(z) dz = 2 \cdot F_{\text{St}}(a) - 1$$

## 7 Expected value: $E(\mathbf{Z}) = 0$

## 8 Variance: $\text{Var}(\mathbf{Z}) = 1$

### Combinatorics

[Introduction](#)[Overview](#)[Summary](#)

### Basics of Probability Theory

[Events, Sample Space & Event Set](#)[Operations with Events](#)[Probability](#)[Rules of Probability](#)[Conditional Probability & Stochastic Independence](#)[Bayes' Rule](#)

### Random variable

[Definition](#)[Distribution Function](#)[Discrete Random Variables](#)[Continuous Random Variables](#)[Expected Value](#)[Variance](#)[Standardization](#)[Quantiles](#)

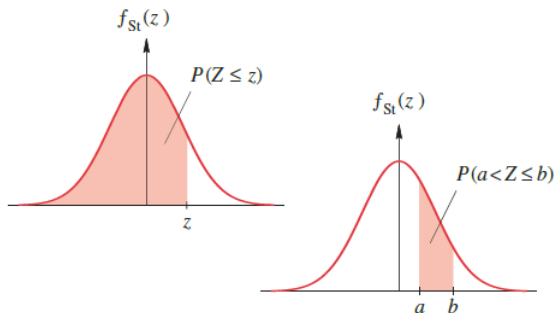
### Stochastic Models & Selected Distributions

[Preface](#)[Discrete Distributions](#)[Continuous Distributions](#)

# Standard Normal Distribution III

## Properties

Probability as area below the density function  $f_{St}(x)$  [Schira, Statistische Methoden der VWL und BWL, 2021]:



### Combinatorics

Introduction

Overview

Summary

### Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

### Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

### Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

## Model

Obtained from the standard normally distributed random variable  $\mathbf{Z}$  by the linear transformation  $\mathbf{X}_N = \mu + \sigma \cdot \mathbf{Z}$  ("de-standardization").

## Normal Distribution

A continuous random variable  $\mathbf{X}_N$  with density function

$$f_N(x) = \frac{1}{\sigma} f_{St} \left( z_x = \frac{x - \mu}{\sigma} \right) = \frac{1}{\sigma \cdot \sqrt{2 \cdot \pi}} e^{-\frac{1}{2} \left( \frac{x - \mu}{\sigma} \right)^2}$$

is called **normally distributed**.

## Properties

- 1 Expected value and variance:

$$\begin{aligned} E(\mathbf{X}_N) &= \mu \\ \text{Var}(\mathbf{X}_N) &= \sigma^2 \end{aligned}$$

- 2 Maximum at  $x = \mu$
- 3 Symmetry w.r.t.  $x \equiv \mu$
- 4 Inflection points at  $x = \mu \pm \sigma$

### Combinatorics

Introduction

Overview

Summary

### Basics of Probability Theory

Events, Sample Space & Event Set

Operations with Events

Probability

Rules of Probability

Conditional Probability & Stochastic Independence

Bayes' Rule

### Random variable

Definition

Distribution Function

Discrete Random Variables

Continuous Random Variables

Expected Value

Variance

Standardization

Quantiles

### Stochastic Models & Selected Distributions

Preface

Discrete Distributions

Continuous Distributions

- 5 Becomes flatter as  $\sigma$  increases:

$$f_N(x) = \frac{1}{\sigma} f_{St} \left( z_x = \frac{x - \mu}{\sigma} \right)$$

- 6 The distribution function

$$F_N(x) = \frac{1}{\sigma \cdot \sqrt{2 \cdot \pi}} \int_{-\infty}^x e^{-\frac{1}{2} \left( \frac{u - \mu}{\sigma} \right)^2} du = F_{St} \left( z_x = \frac{x - \mu}{\sigma} \right)$$

can be reduced to the standard normal distribution  $F_{St}(z)$ .

- 7 Areas:

$$P(\mathbf{X}_N \leq x) = F_{St} \left( z_x = \frac{x - \mu}{\sigma} \right)$$

$$P(a < \mathbf{X}_N \leq b) = F_{St} \left( z_b = \frac{b - \mu}{\sigma} \right) - F_{St} \left( z_a = \frac{a - \mu}{\sigma} \right)$$

$$P(-a < \mathbf{X}_N \leq a) = 2 \cdot F_{St} \left( z_a = \frac{a - \mu}{\sigma} \right) - 1$$

## Combinatorics

[Introduction](#)[Overview](#)[Summary](#)

## Basics of Probability Theory

[Events, Sample Space & Event Set](#)[Operations with Events](#)[Probability](#)[Rules of Probability](#)[Conditional Probability & Stochastic Independence](#)[Bayes' Rule](#)

## Random variable

[Definition](#)[Distribution Function](#)[Discrete Random Variables](#)[Continuous Random Variables](#)[Expected Value](#)[Variance](#)[Standardization](#)[Quantiles](#)

## Stochastic Models & Selected Distributions

[Preface](#)[Discrete Distributions](#)[Continuous Distributions](#)

# Normal Distribution III

## Properties

Density functions  $f_N(x)$  for different Normal distributions [Schira, Statistische Methoden der VWL und BWL, 2021]:

